

AI-Driven Smart Industrial Energy Consumption Optimization Platform

CO1 AT2

Agile Sprint Planning

Identify the Project Vision and Stakeholders

A. The Project Vision: A Deeper Dive

To create a truly compelling Agile project, the vision must bridge the gap between technical execution and business value.

- **The Elevator Pitch:**
 - "To transform traditional, energy-blind manufacturing plants into smart, self-optimizing ecosystems using AI and IoT, driving down costs and carbon footprints simultaneously."
- **Current State vs. Future State:**
 - **Current State:**
 - Factories rely on monthly utility bills to gauge energy use, making it impossible to identify which specific machine or shift caused a spike. Maintenance is reactive, and energy audits are manual and quickly outdated.
 - **Future State:**
 - The AI-Driven Platform acts as the "central nervous system" for the factory's energy grid. It provides continuous, second-by-second visibility, anticipates energy surges before they happen, and dynamically guides operations to be as efficient and green as possible.
- **Core Pillars of the Vision:**
 - **Total Visibility:**
 - Eradicating "blind spots" in industrial energy consumption down to the individual asset level.
 - **Predictive Intelligence:**
 - Shifting from reactive reporting to proactive forecasting using machine learning.
 - **Actionable Optimization:**
 - Providing operators and managers with immediate, data-backed recommendations to reduce waste.
 - **Sustainable Compliance:**
 - Making green manufacturing an effortless byproduct of operational efficiency.

B. Key Stakeholders, Roles, and Motivations

To write effective user stories later, we must deeply understand why each stakeholder cares about this platform. Here is an elaborated breakdown of their pain points and what they value most:

1. Plant Manager (The Optimizer)

- Role: Oversees daily factory operations, production quotas, and facility budget.
- Pain Points: Shrinking profit margins due to rising electricity tariffs; inability to pinpoint why last month's energy bill was 15% higher than expected.
- Value Proposition: They need high-level, real-time dashboards to tie energy costs directly to production output. If a specific production line is inefficient, they want to know immediately to adjust schedules.

2. Machine Operator (The Frontline User)

- Role: Physically operates the heavy machinery (e.g., CNC machines, industrial ovens) on the factory floor.
- Pain Points: Usually disconnected from the financial cost of running their machine. They are judged on speed and output, not energy efficiency.
- Value Proposition: They need simple, non-intrusive visual indicators (like a red/yellow/green gauge) on their workstation screen to let them know if their machine is running optimally, helping them adjust operational parameters without slowing down production.

3. Maintenance Engineer (The Problem Solver)

- Role: Ensures maximum machine uptime and repairs failing equipment.
- Pain Points: Equipment often fails unexpectedly, leading to costly halts in production. Traditional maintenance schedules are often rigid and inefficient.
- Value Proposition: They leverage "energy signatures." A sudden, unexplained spike in a motor's energy draw often means increased friction (e.g., a failing bearing). The platform acts as an early warning system, allowing them to perform predictive maintenance before a catastrophic failure.

4. Sustainability Officer / Energy Auditor (The Compliance Guardian)

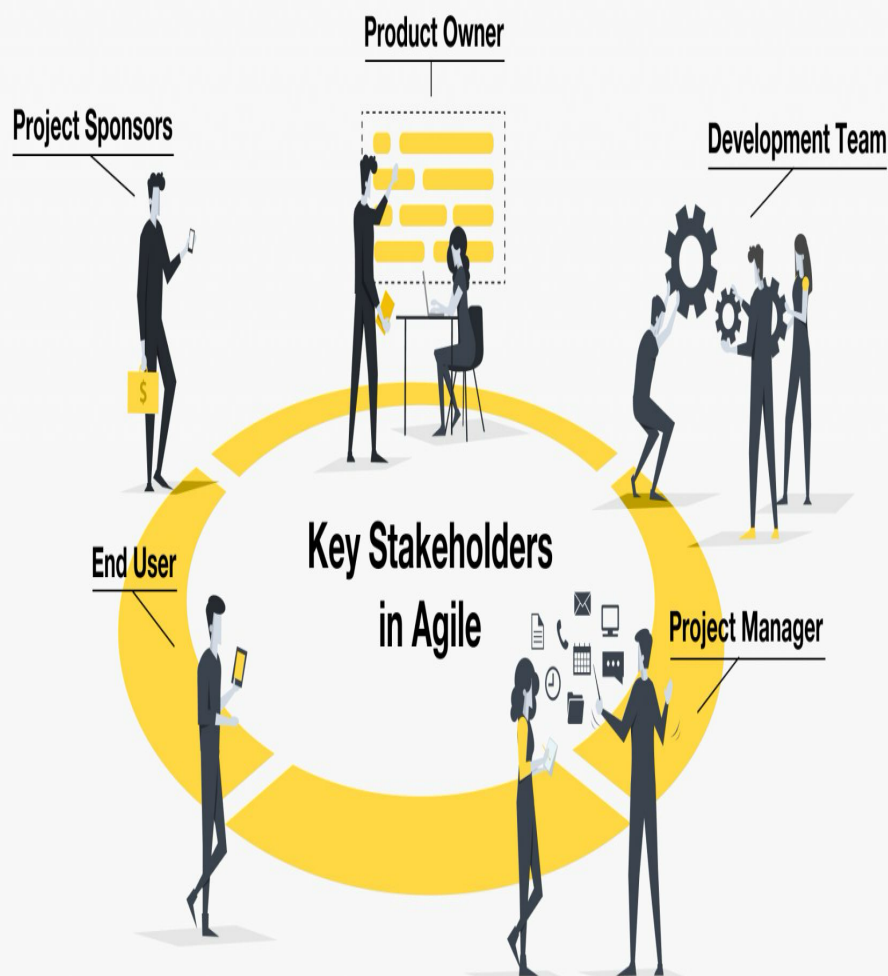
- Role: Ensures the company meets its Environmental, Social, and Governance (ESG) targets and complies with government carbon regulations.
- Pain Points: Gathering data for annual carbon emission reports requires months of manual data entry, estimations, and digging through old utility bills.
- Value Proposition: They require automated, perfectly accurate, audit-ready reports that calculate Scope 1 and Scope 2 emissions based on actual IoT data, saving hundreds of hours of manual work.

5. Executive Sponsor / VP of Operations (The Investor)

- Role: Approves the budget for the AI platform and answers to the board of directors.
- Pain Points: Needs to prove that investing in IoT and AI yields a tangible financial return while boosting the company's public image.
- Value Proposition: They are interested in long-term ROI (Return on Investment). They want high-level summaries showing total dollars saved and total tons of carbon reduced to present to shareholders.

6. IT/OT Integration Manager (The Enabler) [Added for extra depth]

- Role: Manages the factory's network infrastructure (IT) and Operational Technology (OT - like PLCs and SCADA systems).
- Pain Points: Security vulnerabilities introduced by adding hundreds of IoT smart meters to the factory network.
- Value Proposition: Needs the platform to be highly secure, scalable, and capable of handling massive amounts of telemetry data without lagging the factory's core operational network.



Create Epics and User Stories

Epic 1: Real-Time Energy Monitoring

Description: This epic establishes the foundational IoT data pipeline and user interfaces required to capture, process, and display live energy consumption metrics from the factory floor.

- **User Story 1.1 (Core Dashboard):**

- As a Plant Manager, I want to view a centralized real-time dashboard of energy consumption across all machines, so that I can instantly identify areas of excessive energy usage.

- **User Story 1.2 (Operator View):**

- As a Machine Operator, I want to view the live energy status of my assigned machine via a simple color-coded gauge (Green/Yellow/Red), so that I can ensure it is operating within its optimal energy limits without distracting me from production.

- **User Story 1.3 (Granular Filtering):**

- As a Plant Manager, I want to filter the live energy data by specific departments, shifts, or machine types, so that I can isolate inefficiencies in specific areas of the production floor.

- **User Story 1.4 (System Health):**

- As an IT/OT Manager, I want to see connection status indicators for all deployed IoT smart meters, so that I know immediately if a sensor goes offline and stops transmitting data.

Epic 2: AI-Powered Predictive Analytics

Description: This epic focuses on the platform's core intelligence, utilizing historical data and machine learning algorithms to forecast future energy needs and recommend operational optimizations.

- **User Story 2.1 (Demand Forecasting):**

- As a Plant Manager, I want the system to predict peak energy demand for the upcoming week based on inputted production schedules, so that I can plan operations to avoid expensive peak-hour grid tariffs.

- **User Story 2.2 (Prescriptive Optimization):**

- As a Plant Manager, I want the AI to recommend alternative machine scheduling combinations, so that I can maintain production quotas while minimizing total energy costs.

- **User Story 2.3 (Model Confidence):**

- As a Data Scientist, I want to view the confidence interval and historical accuracy metrics of the prediction models, so that I can continually refine and train the AI for better performance.

Epic 3: Automated Anomaly Detection & Alerts

Description: This epic transitions the factory from reactive maintenance to proactive monitoring by using AI to detect deviations from normal energy signatures.

- **User Story 3.1 (Instant Notifications):**

- As a Maintenance Engineer, I want to receive automated SMS and email alerts when a machine exhibits sudden, abnormal energy spikes, so that I can inspect it for potential mechanical faults (e.g., motor friction) before it causes unplanned downtime.

- **User Story 3.2 (Threshold Configuration):**

- As a Plant Manager, I want to configure the sensitivity (percentage deviation) and time duration required to trigger an anomaly alert, so that I can reduce "alert fatigue" from minor, expected power fluctuations.

- **User Story 3.3 (Incident Logging):**

- As a Maintenance Engineer, I want to view a historical log of all triggered energy anomalies alongside their resolution status, so that I can track recurring issues with specific equipment.

Epic 4: Sustainability & Reporting

Description: This epic translates technical energy metrics into business value, providing the necessary documentation for ESG (Environmental, Social, and Governance) compliance and executive oversight.

- **User Story 4.1 (Compliance Reporting):**

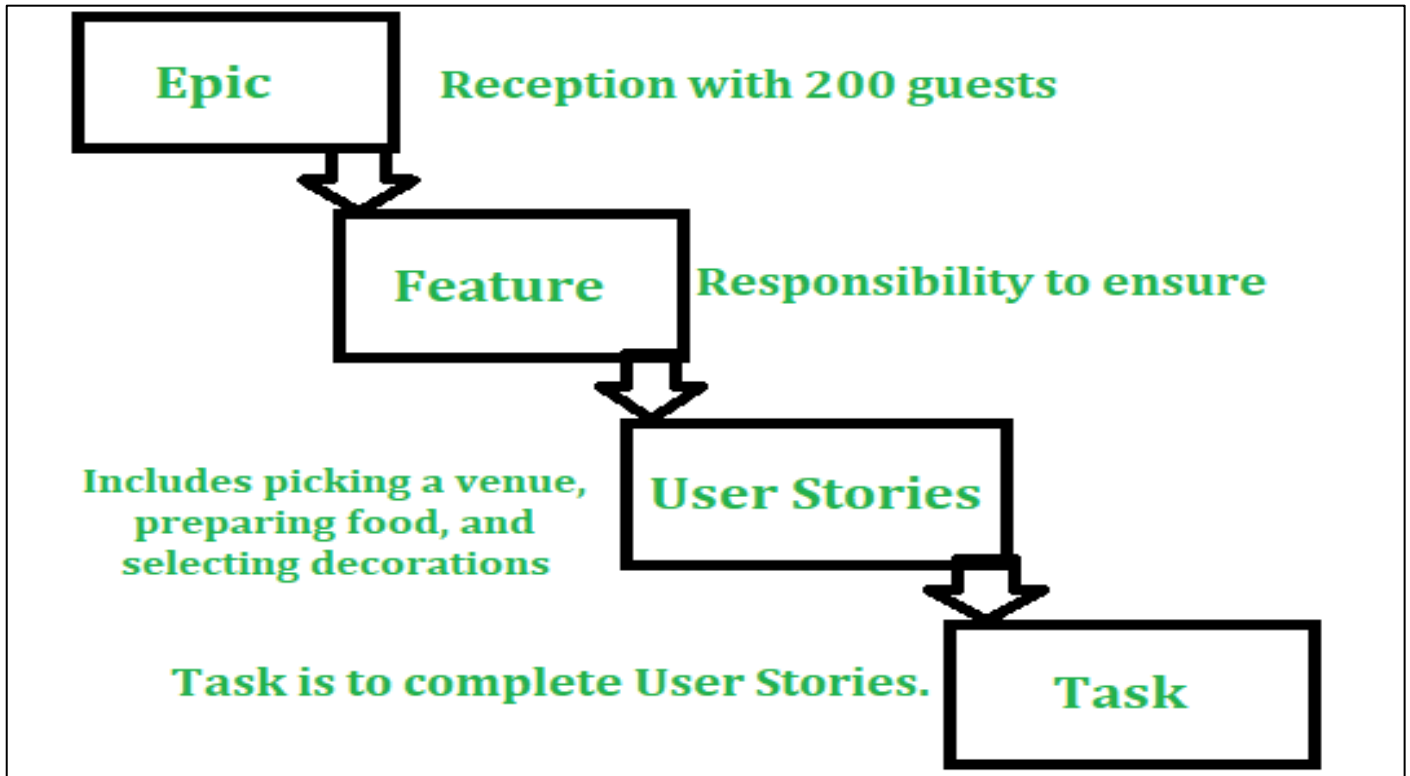
- As a Sustainability Officer, I want to automatically generate monthly carbon emission reports based on consumed energy (Scope 2 emissions), so that I can accurately track our progress toward green manufacturing goals and submit regulatory filings.

- **User Story 4.2 (Historical Comparison):**

- As a Sustainability Officer, I want to compare current monthly energy consumption and emissions against the exact same period from the previous year, so that I can measure the long-term effectiveness of our energy-saving initiatives.

- **User Story 4.3 (Executive Summary):**

- As an Executive Sponsor, I want to view a high-level summary dashboard displaying total monetary savings and total CO2 tonnage reduced since platform launch, so that I can justify the return on investment (ROI) to the board of directors.



Jira Software | Your work | Projects | Filters | Dashboards | People | Apps | Create

Teams In Space | Software project

Easy Agile TeamRhythm | User Story Map | Retrospective

Filter visible issues | Saved Views | Filters / 3 | Layout | Sprint Swimlanes

Book Commercial Space Travel	Manage Commercial Space Travel Booking	Pre-Flight Zero Gravity Training	Know Before You Go Call	Onboard Entertainment	Onboard Accommodation	Onboard Dining
TIS-135	TIS-159	TIS-133	TIS-132	TIS-4	TIS-134	

Sprint 1 | ACTIVE | Customers receive an automated response email when booking | Customers can book onboard dining options | Retrospective | Complete Sprint | 11 | 12 | 0

1st Time Space Tourist / I would like to see suggested travel destinations	1st Time Space Tourist / I want to be able to retrieve my travel booking information	1st Time Space Tourist / I want to understand what the effects Zero Gravity will have on my body	1st Time Space Tourist / I want to know the protocol for onboard emergencies	Returning Space Tourist / I want to listen to my favourite audiobook during my trip to Pluto	1st Time Space Tourist / I would like to compare accommodation	1st Time Space Tourist / I want to be able to book a table in Luna restaurant
FY 22/23 Q1 14/Mar/2019 4:46 AM TIS-100	FY 22/23 Q1 06/Jun/2022 11:15 AM TIS-160	FY 22/23 Q1 28/Mar/2019 3:32 AM TIS-136	FY 22/23 Q1 28/Mar/2019 3:32 AM TIS-137	FY 22/23 Q2 28/Mar/2019 3:39 AM TIS-143	FY 22/23 Q2 06/Jun/2022 11:49 AM TIS-173	FY 22/23 Q3 06/Jun/2022 11:58 AM TIS-17

Update LodgingController to handle multiple travel providers in one reservation
None
14/Mar/2019 4:46 AM
TIS-97

Travel credit codes are failing to be recognised
None
06/Jun/2022 11:16 AM
TIS-162

Implement 'Resy' booking system for a dining options
None
06/Jun/2022 12:00 PM
TIS-17

You're in a company-managed project

Acceptance Criteria

US 1.1: Real-Time Dashboard

Positive Scenario (The Happy Path)

- Given the IoT sensors are actively transmitting data and the Facility Manager is logged in,
- When the user navigates to the "Live Monitoring" tab,
- Then the dashboard must display current kWh usage per machine, updating every 5 seconds without requiring a manual page refresh.

Negative Scenario (Data Interruption)

- Given a machine's IoT sensor loses network connection or power,
- When the dashboard attempts its 5-second data pull,
- Then the UI must change that specific machine's status indicator to "Offline" (e.g., a red icon) after 15 seconds of missing data.

Non-Functional & UI Constraints

- Given the "Live Monitoring" dashboard is rendering,
- When the page loads,
- Then the initial data fetch and UI rendering must complete within 2 seconds.
- Given the data is displayed,
- When viewing the machine grid,
- Then the data points must clearly show Machine ID, Current Status (Active/Idle/Offline), and Current Energy Draw (kWh).

US 1.2: Automated Alerts

Positive Scenario (Threshold Breach)

- Given a machine is categorized under active production mode,
- When its energy draw continuously exceeds the established baseline by 15% for more than 2 minutes,
- Then the system must immediately dispatch a push notification and an SMS alert to the assigned Machine Operator.

Edge Case (Alert Fatigue / Cooldown)

- Given an alert has just been triggered and sent to the operator,
- When the machine's energy draw fluctuates above and below the 15% threshold rapidly,
- Then the system must initiate a 15-minute "cooldown" period where no duplicate SMS alerts are sent for that specific machine to prevent spam.

Data & Content Constraints

- Given the system is generating the automated alert,
- When the SMS and push notification are formatted,
- Then the message body must include: the Timestamp, Machine ID, Baseline kW, and the recorded Spiked kW.

US 2.1: Predictive Forecasts

Positive Scenario (Successful Prediction)

- Given the AI model has successfully ingested and processed at least 30 days of historical energy data,
- When the Production Supervisor views the "Demand Forecast" module,
- Then the system should display a 7-day interactive predictive graph of energy demand.

Negative Scenario (Insufficient Data)

- Given a newly added machine has only been monitored for 10 days,
- When the AI engine attempts to generate a forecast for this specific machine,
- Then the UI must display a disabled graph with a warning banner reading: "Insufficient historical data. Minimum 30 days required for forecasting."

Non-Functional & Analytical Constraints

- Given the 7-day predictive graph is displayed,
- When the user hovers over a future data point,
- Then a tooltip must show the predicted peak kW, alongside a confidence interval (e.g., $\pm 5\%$) indicating the model's accuracy minimum of 85%.

Given



Existing conditions or prerequisites that need to be in place for a particular scenario to occur.

When



The specific action or input that you want to observe in order to see how the system behaves.

Then



Expected outcome after the trigger event has occurred.

Prioritize the Product Backlog

The backlog is prioritized based on business value, focusing on foundational data visibility first, followed by advanced AI optimization, and finally compliance reporting.

Priority	Epic	User Story	Business Value	Dependency
1 (High)	1. Monitoring	US 1.1: Real-time Dashboard	Core foundation; enables immediate visibility.	Hardware/Sensor Setup
2 (High)	1. Monitoring	US 1.2: Automated Alerts	Prevents critical waste/faults.	US 1.1 (Data pipeline)
3 (Med)	2. Prediction	US 2.1: Peak Demand Forecast	Drives cost-savings via load shifting.	Historical data collection
4 (Med)	2. Prediction	US 2.2: Utilization Recommendations	Optimizes overall plant efficiency.	US 2.1 (AI baseline)
5 (Low)	3. Reporting	US 3.1: Carbon Emission Reports	Important for compliance, but not operational.	US 1.1 (Accurate data)

Plan the Sprint Backlog

Task 1 (Backend): Time-Series Database Configuration

- **Schema Design:**
 - Define the exact data structure for the time-series database (e.g., InfluxDB), clearly separating tags (indexed metadata like Machine_ID, Facility_Zone, Machine_Type) from fields (the actual recorded metrics like Current_kWh, Voltage, Amperage).
- **Data Retention Policies:**
 - Configure data lifecycle rules to automatically downsample high-frequency data (e.g., rolling up 5-second interval data into 1-hour averages after 7 days) to optimize database performance and storage costs.
- **Infrastructure Provisioning:**
 - Set up the database environment securely using Docker containers or a managed cloud service, establishing the initial administrative credentials and access tokens.

Task 2 (Data Engineering): MQTT Ingestion Pipeline

- **Broker Setup:**
 - Deploy and configure an MQTT broker (such as Eclipse Mosquitto or AWS IoT Core) to handle incoming sensor traffic efficiently. Topic Architecture: Establish a logical, hierarchical MQTT topic structure (e.g., plant_1/assembly_line/machine_042/energy) to easily route, filter, and debug incoming messages.
- **Consumer Service:**
 - Write a lightweight backend service (in Python or Node.js) that subscribes to the MQTT broker, validates the incoming JSON payload structure, and writes the parsed data points directly into the database.

Task 3 (Backend API): RESTful Data Endpoints

- **Endpoint Creation:**
 - Develop specific API routes, such as a GET /api/v1/energy/live (for a grid view of all machines) and GET /api/v1/energy/live/{machine_id} (for detailed single-machine views), optimized to query the most recent data points rapidly.

- **Error Handling:**

- Build robust HTTP error responses (e.g., returning a 404 if a queried machine ID does not exist, or a 503 if the database connection drops) to pass meaningful context to the frontend UI.

Task 4 (Frontend): UI & Dashboard Development

- **Component Design:**

- Build reusable UI components using a modern framework (like React or Vue), specifically creating a "Machine Status Card" (showing Active/Idle/Offline states) and a central layout grid.

- **Data Fetching Logic:**

- Implement a reliable fetching mechanism to pull new data from the API every 5 seconds.

- **State Management:**

- Configure the frontend state to gracefully handle data interruptions, adhering to the acceptance criteria by switching a machine's visual indicator to "Offline" (red) if the data fetch returns null or fails for more than 15 seconds.

- **Visualization:**

- Integrate a performant charting library (such as Chart.js, Recharts, or Grafana panels) that is optimized for rendering real-time time-series data without causing memory leaks over prolonged dashboard viewing sessions.

Task 5 (QA/Testing): Quality Assurance & Integration

- **Data Simulation:**

- Create a mock data generator script that mimics IoT hardware, rapidly publishing randomized but realistic energy metrics to the MQTT broker to test the pipeline without waiting for physical factory sensors to be installed.

- **Latency Testing:**

- Monitor the timestamp of a simulated sensor payload from the moment of ingestion to frontend rendering, verifying that the end-to-end system latency remains strictly under the 5-second acceptance criteria.

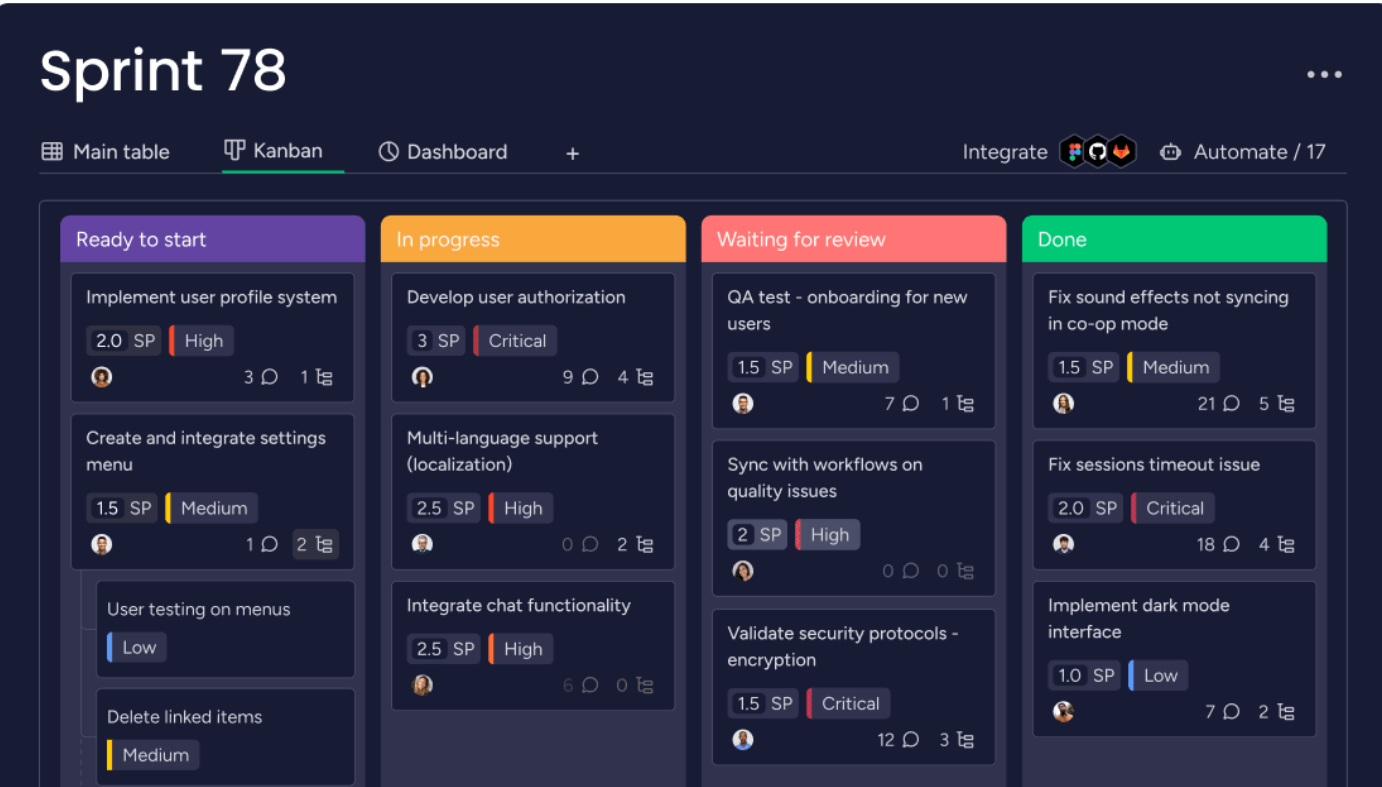
- **API Load Testing:**

- Use tools like Postman or JMeter to simulate multiple dashboard instances pinging the API simultaneously, ensuring the backend can handle the continuous read volume without degrading performance.

Estimate Effort Using Story Points

Estimations are made using the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21) during a Planning Poker session.

USER STORY	STORY POINTS	JUSTIFICATION
US 1.1 (DASHBOARD)	8	High complexity. Requires establishing the initial full-stack architecture (database setup, MQTT pipeline, API, and UI).
US 1.2 (ALERTS)	5	Medium complexity. The data pipeline is already built in US 1.1, but requires implementing logic thresholds and a notification gateway (SMS/Push).
US 2.1 (FORECASTING)	13	Very high complexity. Requires data cleaning, feature engineering, and training a machine learning model (e.g., LSTM), plus UI integration for forecasting graphs.
US 3.1 (REPORTING)	3	Low complexity. Involves querying existing data and formatting it into a downloadable PDF/CSV template.



Present the Sprint Goal and Expected Deliverables

Sprint 1 Goal:

“Establish a secure, end-to-end data pipeline from factory floor sensors to the cloud, and deliver a functional real-time energy monitoring dashboard for at least one primary manufacturing machine.”

Strategic Justification of the Goal:

This goal was selected because it tackles the highest-risk technical dependency first (the hardware-to-cloud data pipeline) while simultaneously delivering immediate, visible business value to the Facility Manager. By validating that data can be ingested securely and visualized with minimal latency, the team establishes a solid architectural foundation for the subsequent AI and predictive forecasting features planned for future sprints.

Elaborated Expected Sprint Deliverables:

Deployed Time-Series Database & Ingestion Pipeline:

- A fully configured time-series database instance (e.g., InfluxDB or TimescaleDB) running in a staging environment.
- A secure MQTT broker actively receiving and authenticating simulated IoT sensor data payloads.
- Verifiable data tables that correctly parse and store machine ID, timestamps, and energy metrics (kW/h, voltage, amperage).

Live Monitoring Web Dashboard (MVP):

- A secure, web-based frontend application accessible by the Facility Manager via browser.
- A primary UI dashboard featuring dynamic line charts (using a library like Chart.js or Grafana) that render incoming time-series data.
- An auto-refresh mechanism that successfully pulls new data points and updates the visual graphs every 5 seconds without requiring a manual browser reload.
- A basic "Machine Status" indicator that visually alerts the user (e.g., turns red) if the data connection drops.

API Endpoint Documentation (Swagger/OpenAPI):

- Fully documented and tested RESTful API endpoints used to connect the frontend to the database.

- Documentation will include clear request parameters, authorization header requirements (JWT), and standard JSON response schemas for both successful data retrieval and error handling (e.g., 404 Machine Not Found).

Interactive Sprint Review Demonstration:

- A live, end-to-end demonstration scheduled with key stakeholders (Facility Manager, Product Owner).
- The demo will feature a live simulation of a machine turning on, generating energy data, and that data appearing on the dashboard within the 5-second latency tolerance.
- Collection of stakeholder feedback to inform the backlog grooming for Sprint 2.

Definition of Done (DoD) for Sprint 1:

All deliverables will meet the following criteria before being marked complete:

- Code is version-controlled, peer-reviewed, and merged into the main staging branch.
- Unit and integration tests for the API and data pipeline are passing.
- Acceptance criteria for User Story 1.1 are verified by the QA lead.
- The system is deployed and accessible in the staging environment.

